

SRM UNIVERSITY ANDHRA PRADESH
Department of Electrical and Electronics Engineering
CLASSROOM LEARNING ASSESSMENT-I
EEE 310 – Principles of Signal Processing (Semester 5/Fall 2026)

Duration: 45 Minutes**Maximum Marks:** 25**Date:** _____**Roll No.:** _____**Course Outcomes**

- CO1:** Analyze fundamental principles of signals and systems and apply them to physical systems in the time and frequency domains.
- CO2:** Apply Fourier-series concepts to modern digital signal processing, harmonic analysis in power electronics, and power-system analysis/stability.
- CO3:** Apply Fourier and Laplace transforms for frequency-domain system analysis, prediction of system behavior/stability, and control-system design.
- CO4:** Explain the need for discretization of continuous-time signals and analyze reconstruction of the original continuous-time signal from its samples.

Instructions

1. Attempt **all** questions. Section A carries 10 marks and Section B carries 15 marks.
2. In Section A, a bare “True” or “False” receives **zero credit**. Every answer must be justified by an equation, derivation, counterexample, or exact calculation.
3. Clearly mark all breakpoints, amplitudes, time reversal/scaling/shifting, and axes in sketches.
4. Unless otherwise stated, use $u(t) = 1$ for $t \geq 0$, $u(t) = 0$ for $t < 0$, $r(t) = t u(t)$, and $\text{rect}(t) = 1$ for $|t| < \frac{1}{2}$, 0 for $|t| > \frac{1}{2}$; values exactly at the jump points are immaterial unless explicitly specified.

SECTION A (5 × 2 = 10 Marks)

Q. No.	Question	Marks	CO	BL
1	True or False? Let $x_1(t) = \cos(4\pi t), \quad x_2(t) = \sin(6\pi t),$ and define $z(t) = x_1(2t - 1) + x_2(3t).$ The fundamental period of $z(t)$ is $T_0 = \frac{1}{36}$ s. Determine the truth value by deriving the individual transformed periods and the <i>smallest</i> common period.	2	CO1	BL4
2	True or False? For every finite-energy signal $x(t)$, if $x_e(t) = \frac{x(t) + x(-t)}{2}, \quad x_o(t) = \frac{x(t) - x(-t)}{2},$ then $x_e(t)$ and $x_o(t)$ are orthogonal and $E_x = E_{x_e} + E_{x_o}$. Establish or disprove the statement using the energy inner product; do not use a verbal symmetry argument alone.	2	CO1	BL5
3	True or False? A nonzero periodic signal having finite, nonzero energy in one fundamental period can never be an energy signal; its total energy is infinite but its average power is finite. Also, every finite-energy aperiodic signal has zero average power. Verify both clauses directly from the definitions of E and P.	2	CO1	BL5

Q. No.	Question	Marks	CO	BL
4	True or False? If a periodic signal satisfies half-wave symmetry $x\left(t + \frac{T_0}{2}\right) = -x(t),$ then (i) its average value over one period is zero, and (ii) its RMS value computed over <i>any</i> interval of length $T_0/2$ is identical to its RMS value over a full period. Prove or disprove both parts analytically.	2	CO1	BL4
5	True or False? Given $r(t+1) - 2r(t) + r(t-1) = \text{rect}(\frac{t}{2})$. Decide by evaluating the left-hand side piecewise. Then obtain its first and second distributional derivatives in terms of $u(t)$ and $\delta(t)$, respectively.	2	CO1	BL4

SECTION B (3 × 5 = 15 Marks)

Q. No.	Question	Marks	CO	BL
6	Consider $x(t) = \begin{cases} 0, & t < -2, \\ 1, & -2 \leq t < 0, \\ 2, & 0 \leq t < 1, \\ 0, & t \geq 1. \end{cases}$ (a) For $y(t) = 2 - 3x(1 - 2t),$ identify the exact sequence of time reversal, time scaling, time shifting, amplitude scaling/reversal, and amplitude shifting. Determine all transformed breakpoints and amplitudes and sketch $y(t)$. (3 marks) (b) Obtain $x_e(t)$ and $x_o(t)$ <i>piecewise</i>, and sketch both on the same time axis. Verify analytically that $x(t) = x_e(t) + x_o(t)$. (2 marks)	5	CO1	BL4
7	A waveform has fundamental period $T_0 = 4$ ms. Over the first half-period, $x(t) = \begin{cases} A \frac{t}{1 \text{ ms}}, & 0 \leq t < 1 \text{ ms}, \\ A, & 1 \text{ ms} \leq t < 2 \text{ ms}, \end{cases}$ and for all t it satisfies $x(t + 2 \text{ ms}) = -x(t).$ Take $A = 4$. (a) Derive the average value, RMS value, average power, and total-energy classification of $x(t)$. Do not integrate over the full period if symmetry can be used rigorously. (2 marks) (b) Define $z(t) = x(t) + 2 \cos(2\pi 500t) + 3 \sin(2\pi 1000t).$ Using half-wave symmetry, prove by integration that each added sinusoid is orthogonal to $x(t)$ over T_0. Hence determine the average power and fundamental period of $z(t)$. (3 marks)	5	CO1	BL5

Q. No.	Question	Marks	CO	BL
8	Let $x_a(t) = e^{-a t }, \quad a > 0,$ and let $p(t) = \text{rect}\left(\frac{t}{2}\right), \quad \phi_1(t) = p(t), \quad \phi_2(t) = \text{sgn}(t)p(t).$ (a) Show that $x_a(t) \rightarrow 1$ pointwise as $a \rightarrow 0^+$. For every $a > 0$, calculate E_{x_a} and P_{x_a}, and compare these with the energy/power of the limiting DC signal $x(t) = 1$. Explain the apparent change of signal class using only the limiting expressions obtained. Further determine the energy contained in $[-T, T]$ and evaluate its limiting behavior first as $T \rightarrow \infty$ for fixed $a > 0$, and then as $a \rightarrow 0^+$. (3 marks) (b) Prove that ϕ_1 and ϕ_2 are orthogonal and, without expanding the square directly, compute the energy of $q(t) = 3\phi_1(t) - 4\phi_2(t).$ (1 mark) (c) Express $p(t)$ exactly using unit-step functions and obtain $\frac{dp(t)}{dt}$ in terms of shifted impulses. (1 mark)	5	CO1	BL5

— End of Question Paper —